Your Code

```python
names = [
" Arun Kumar ",
"ARUN KUMAR",
" arun kumar",
"Bala Krishnan",
"BALA KRISHNAN ",
"Charan",
" charan ",
"DIVYA DEVI",
"Divya Devi"
]

normalized = [name.strip().lower() for name in names]
print("Normalized:")
print(normalized)

unique_customers = []
for name in normalized:
    if name not in unique_customers:
        unique_customers.append(name)
unique_customers.sort()

print("\nUnique Customers:")
```

Input

stdin input

Output

```
[' arun kumar ', 'bala krishnan',
'charan', 'divya devi']

Most Frequent Customer:
arun kumar
```

```python
products = [
    ("Laptop", 5, 55000),
    ("Mouse", 25, 700),
    ("Keyboard", 8, 1200),
    ("Monitor", 12, 15000),
    ("Printer", 4, 12000),
    ("Headset", 15, 2500)
]

print("Inventory Values:")
inventory_values = []
for name, qty, price in products:
    value = qty * price
    inventory_values.append((name, value))
    print(f"{name} : {value}")

total_value = sum(v for _, v in inventory_values)
print(f"\nTotal Inventory Value: {total_value}")

print("\nLow Stock:")
for name, qty, price in products:
    if qty < 10:
        print(name)

```

stdin input

Output

```
Inventory Values:
Laptop : 275000
Mouse : 17500
Keyboard : 9600
Monitor : 180000
Printer : 48000
```

```python
import string

feedback = """
Python programming is easy to learn.
Python programming is powerful.
Learning Python requires practice.
Practice makes programming skills better.
"""

text = feedback.lower()
text = text.translate(str.maketrans('', '', string.punctuation))
words = text.split()

total_words = len(words)
unique_words = set(words)
num_unique = len(unique_words)

freq_python = words.count("python")
freq_programming = words.count("programming")

repeated_words = [word for word in unique_words if words.count(word) >

longest_word = max(unique_words, key=len)

```

stdin input

Output

```
Total Words: 19
Unique Words: 13

Frequency of python: 3
Frequency of programming: 3
```

Your Code

```python
employees = [
    ("E101", "Arun", ["Python", "SQL", "ML"]),
    ("E102", "Bala", ["Java", "SQL", "HTML"]),
    ("E103", "Charan", ["Python", "ML", "SQL"]),
    ("E104", "Divya", ["Python", "Java", "Cloud"]),
    ("E105", "Esha", ["Python", "ML", "Cloud"])
]

python_emp = [name for _, name, skills in employees if "Python" in skil
print("Python:")
print(", ".join(python_emp))
print()

python_ml = [name for _, name, skills in employees if "Python" in skill
print("Python + ML:")
print(", ".join(python_ml))
print()

sql_not_java = [name for _, name, skills in employees if "SQL" in skill
print("SQL but not Java:")
print(", ".join(sql_not_java))
print()
```

Input

stdin input

Output

```
Python:
Arun, Charan, Divya, Esha

Python + ML:
Arun, Charan, Esha
```

Your Code

```python
transactions = [
    ("T101", "Deposit", 5000),
    ("T102", "Withdraw", 2000),
    ("T103", "Deposit", 8000),
    ("T104", "Withdraw", 1500),
    ("T105", "Deposit", 3000),
    ("T106", "Withdraw", 7000)
]

total_deposit = sum(a for _, t, a in transactions if t == "Deposit")
total_withdraw = sum(a for _, t, a in transactions if t == "Withdraw")
net_balance = total_deposit - total_withdraw

largest_deposit = max(a for _, t, a in transactions if t == "Deposit")
largest_withdraw = max(a for _, t, a in transactions if t == "Withdraw"

greater_than_4000 = [tid for tid, _, a in transactions if a > 4000]

deposit_count = sum(1 for _, t, _ in transactions if t == "Deposit")
withdraw_count = sum(1 for _, t, _ in transactions if t == "Withdraw")

avg_transaction = sum(a for _, _, a in transactions) / len(transactions
```

Input

stdin input

Output

```
Total Deposit: 16000
Total Withdrawal: 10500
Net Balance: 5500

Largest Deposit: 8000
Largest Withdrawal: 7000
```

Your Code

```python
passwords = [
    "Python@123",
    "python123",
    "PYTHON@123",
    "Py@45",
    "Python123",
    "Java#4567",
    "Admin@2026"
]

valid_passwords = []
invalid_passwords = []

for p in passwords:
    errors = []
    if len(p) < 8:
        errors.append("Less than 8 characters")
    if not any(c.isupper() for c in p):
        errors.append("No uppercase")
    if not any(c.islower() for c in p):
        errors.append("No lowercase")
    if not any(c.isdigit() for c in p):
        errors.append("No digit")
```

Input

```
stdin input
```

Output

```
Valid:
Python@123
Java#4567
Admin@2026

Invalid:
```

9. Analyze what happens if record[0] is changed instead.

10. Explain why modifying record[0] directly is not allowed.

Expected Output:

['arun', 'bala', 'charan']

Your Code

```python
records = [
    (" Arun ", "Python"),
    ("ARUN", "Python"),
    (" Bala ", "Java"),
    ("bala", "Java"),
    ("Charan", "Python")
]

names = []

for record in records:
    name = record[0].strip().lower()
    if name not in names:
        names.append(name)

print(names)
```

Input

stdin input

Output

```
['arun', 'bala', 'charan']
```

Your Code

```python
from collections import defaultdict

students = [
    (" A001 ", "ARUN KUMAR", [78, 85, 92]),
    ("A002", "Bala", [45, 67, 52]),
    (" a003", "arun kumar", [88, 91, 95]),
    ("A004", "CHARAN", [72, 76, 81]),
    ("A005", "Divya ", [55, 48, 62])
]

normalized_records = []
totals = defaultdict(int)
averages = {}
passed_all = []
failed_at_least_one = []
name_counts = defaultdict(int)

for sid, name, marks in students:
    sid_n = sid.strip().upper()
    name_n = name.strip().lower()
    normalized_records.append((sid_n, name_n, marks))
```

Input

stdin input

Output

```
Normalized Records:
A001 arun kumar [78, 85, 92]
A002 bala [45, 67, 52]
A003 arun kumar [88, 91, 95]
A004 charan [72, 76, 81]
A005 divya [55, 48, 62]
```

Your Code

```python
students = [
    ("Arun", [78, 85, 92, 67]),
    ("Bala", [45, 67, 55, 49]),
    ("Charan", [90, 92, 88, 95]),
    ("Divya", [65, 72, 70, 68]),
    ("Esha", [88, 76, 91, 84])
]

totals = {}
averages = {}

print("Student Totals:")
for name, marks in students:
    total = sum(marks)
    totals[name] = total
    print(f"{name} : {total}")

print("\nStudent Averages:")
for name, marks in students:
    avg = sum(marks) / len(marks)
```

Input

stdin input

Output

```
Student Totals:
Arun : 322
Bala : 216
Charan : 365
Divya : 275
Esha : 339
```